Padraig Lavery

Software Engineer – Backend, Cloud & Production Reliability | Node.js, .NET/C#, AWS, Kubernetes

Belfast, Northern Ireland | LinkedIn | GitHub

Looking for: senior or mid-senior roles in backend, platform/DevOps, reliability or security engineering, and forward deployed or applied AI roles. Belfast hybrid, remote within Northern Ireland or the UK, or Dublin.

PROFILE

Software engineer with 4+ years building and running high-volume, security-sensitive systems for UK and Middle East banking clients (Angular, .NET/C#, Node.js, PostgreSQL on AWS and Kubernetes). Go-to engineer for production incidents, focused on fixing root causes and leaving runbooks behind so they don't recur. Leads projects and team retrospectives, led a cross-team test-automation and CI initiative, and has mentored four new starters. Building applied LLM skills through Incident Copilot, an incident-triage service using RAG and agentic tool-calling.

CORE STRENGTHS

- **Backend and APIs:** Node.js, C#/.NET Core, Python (FastAPI), microservices, OpenAPI, PostgreSQL, MongoDB
- **Cloud and platform:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **Reliability:** incident ownership, root cause analysis, safe retries and timeouts, runbooks and SOPs
- **Security and compliance:** PCI DSS vulnerability remediation, Snyk, OWASP, audit logging, input validation
- **Front end:** Angular, HTML/CSS, WCAG accessibility
- **Applied AI:** retrieval-augmented generation, agentic tool-calling, pgvector, local LLMs (Ollama), AWS Bedrock and SageMaker

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Production reliability and incident ownership

- Eliminated recurring daily out-of-memory crashes in a high-volume transaction ingestion service by re-architecting cache loading to run once per batch instead of per file; diagnosed and mitigated further memory-exhaustion failures in ETL jobs.
- Resolved a live customer-facing outage caused by malformed load-balancer group naming in a Kubernetes/Istio ingress setup, and permanently corrected the Helm deployment templates so it could not recur.
- Redesigned a month-end transactional invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation, so failures on the highest-volume partners no longer abort the whole run.
- Authored runbooks and SOPs enabling the production services team to resolve recurring issues without engineering escalation.

Security, compliance and platform engineering

- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals (agent, customer, timestamp, session, IP) to meet client security and compliance audits.
- Hardened authentication and account services: removed stack traces and SQL details from error responses and standardised input validation across partner microservices.
- Authored the engineering migration guide for AWS SDK for JavaScript v2 to v3 across Lambdas, Fargate services and microservices, and planned testing for the SDK and Node 22 Lambda runtime upgrades across repositories and bank deployments, remediating critical Snyk-flagged vulnerabilities for PCI DSS.
- Audited 100+ S3 buckets, applying lifecycle policies and removing obsolete resources; automated scheduled file processing with EventBridge, Lambda and Fargate.

Features, delivery and leadership

- Led the technical migration of email templates onto dedicated brand domains across four banking brands (MJML, .NET Core), preserving deliverability.
- Built full stack features across web and mobile, including a secure e-code resend flow and targeted redemption pop-ups; fixed WCAG colour-contrast failures (1.7:1 to 4.5:1).
- Led a cross-team initiative for automated API and feature testing, including OpenAPI documentation, unit tests and stabilised GitHub Actions CI.
- Built the CI/CD pipeline for a UI Builder service in an internal AI investigation project.
- Mentored and onboarded four new starters; run the team's retrospectives and author technical guides.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a web-based API for calculating cloud costs in a small agile team, using Go, Docker, Kubernetes and Git.

PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Incident-triage service that retrieves runbook content and suggests likely root causes and next steps, with agentic tool-calling for live service status. Moved from AWS Bedrock to local models via Ollama to remove inference costs.

AI-Based Credit Card Fraud Detection (final year project) | Angular, Flask, GraphQL, Docker, Kubernetes, AWS SageMaker

- End-to-end web app flagging irregular credit card transactions, with a model served via SageMaker.

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026